Section 5–1 Basic Combinational Logic Circuits:

4. Write the output expression for each circuit as it appears in Figure 5–56 and then change each circuit to an equivalent AND-OR configuration. (d)

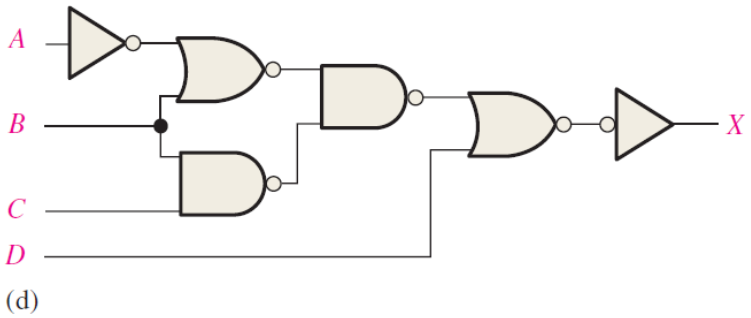


FIGURE 5–56

Section 5–2 Implementing Combinational Logic:

13. Implement a logic circuit for the truth table in Table 5–8.

A	*B*	*C*	*X*
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

19. Minimize the gates required to implement the function of the circuit in each part of Figure 5–56 in SOP form. (d)

Section 5–3 The Universal Property of NAND and NOR Gates:

21. Implement the logic circuit in Figure 5–58 using only NAND gates.

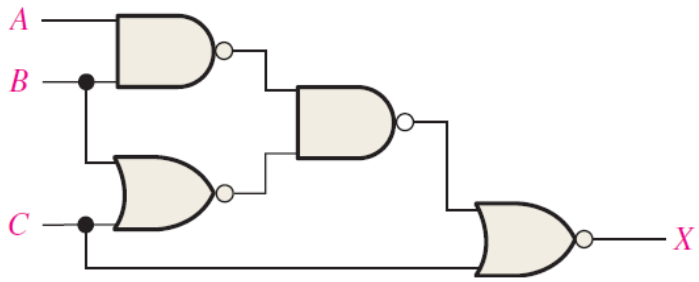


FIGURE 5–58

23. Repeat Problem 21 using only NOR gates.

Section 5–5 Pulse Waveform Operation:

28. Given the logic circuit and the input waveforms in Figure 5–59, draw the output waveform.

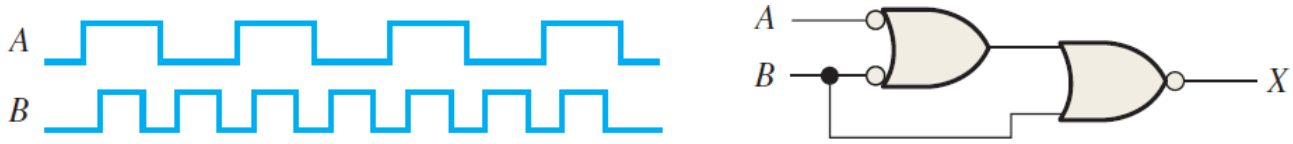


FIGURE 5–59

31. For the input waveforms in Figure 5–62, what logic circuit will generate the output waveform shown?

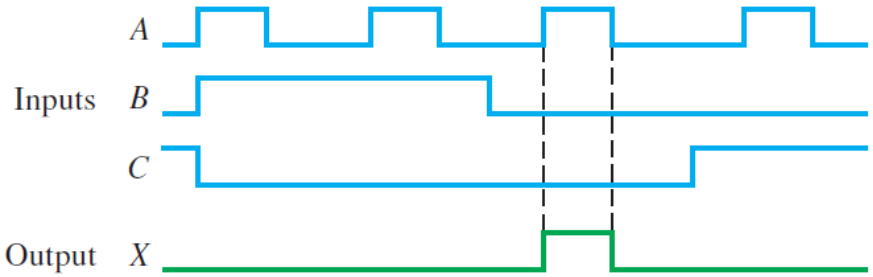


FIGURE 5–62

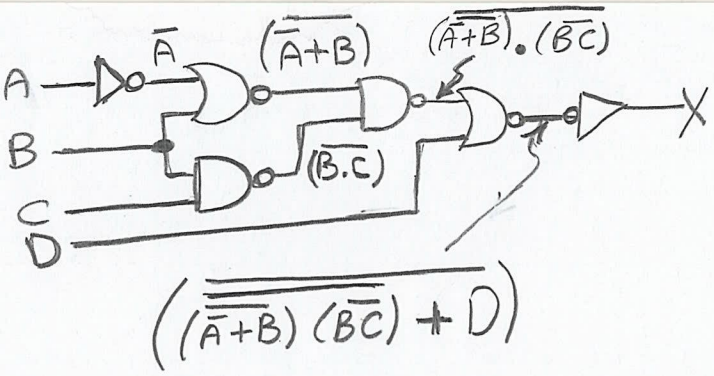
Special Design Problems:

56. Design a logic circuit to produce a HIGH output only if the input, represented by a 4-bit binary number, is greater than twelve or less than three. First develop the truth table and then draw the logic diagram.

Ch5

4 (d)

& 19



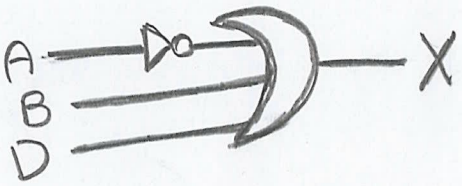
$$X = \left(\overline{\overline{(\overline{\bar{A}+B})(\overline{BC})}+D}\right)$$

$$= \overline{(\overline{\bar{A}+B})(\overline{BC})}+D$$

$$= \overline{(\overline{\bar{A}+B})} + \overline{(\overline{BC})} + D$$

$$= \bar{A}+B+BC+D$$

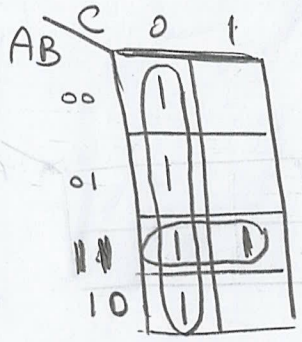
$$= \bar{A}+B+D \Rightarrow$$



13

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

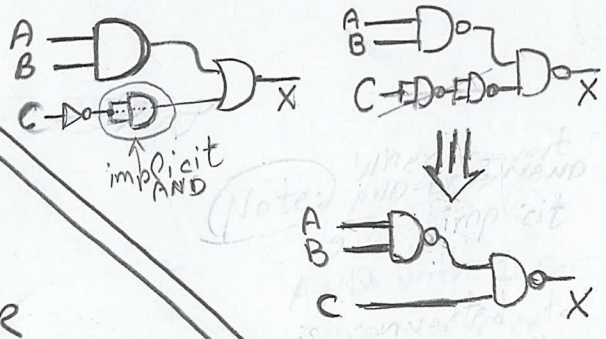
To get SOP ⟹



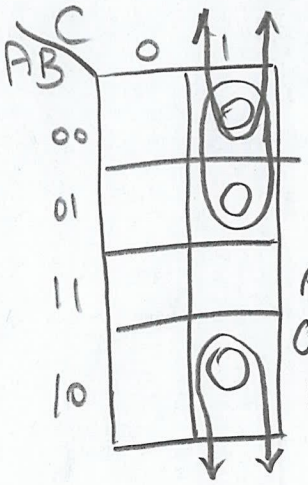
$$X = \bar{C} + AB$$

AND-OR

NAND



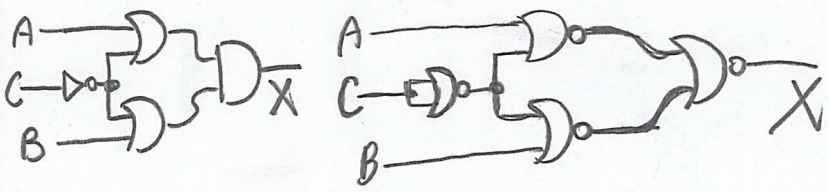
To get POS ⇓



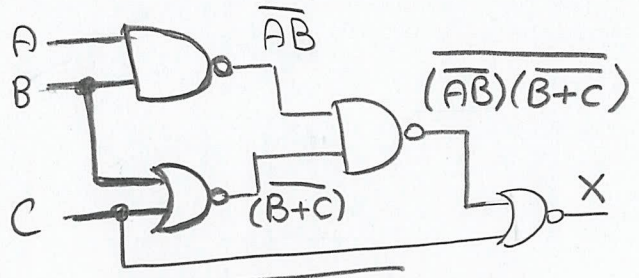
$$X = (\bar{C}+A)(\bar{C}+B)$$

AND-OR

NOR



21 & 23



$$\Rightarrow X = \overline{\left(\overline{(\overline{AB})(\overline{B+C})} + C\right)}$$

$$X = \overline{\overline{(\overline{AB})(\overline{B+C})}\,\overline{C}}$$

$$= \overline{AB}\,(\overline{B+C})\,\overline{C}$$

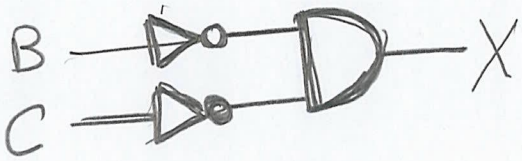
$$= (\overline{A}+\overline{B})\,\overline{B}\,\overline{C}\,\cancel{\overline{C}}$$

$$= \overline{A}\,\overline{B}\,\overline{C} + \overline{B}\,\cancel{\overline{B}}\,\overline{C}$$

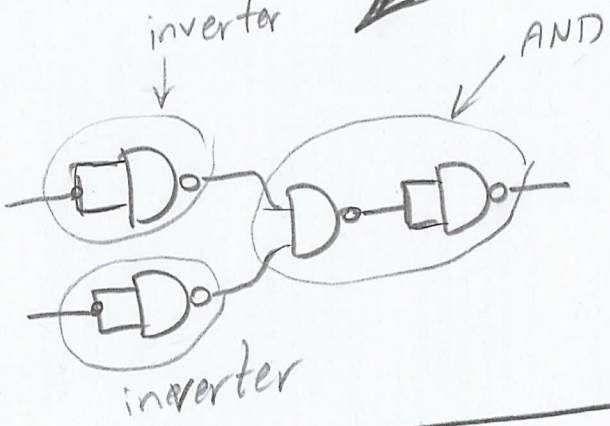
$$= (\overline{A}+1)\,\overline{B}\,\overline{C}$$

$$= \overline{B}\,\overline{C}$$

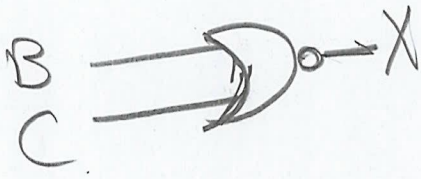
$\Longrightarrow$



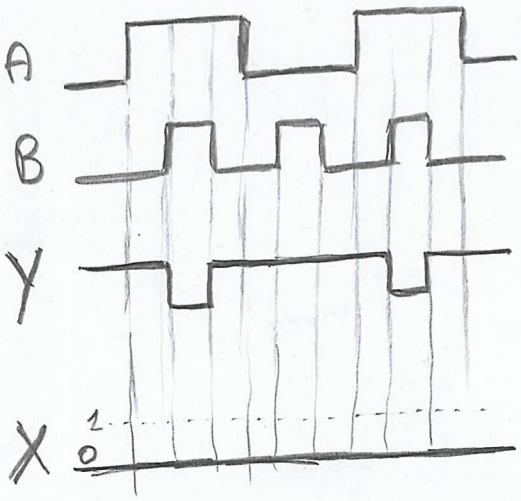
NAND

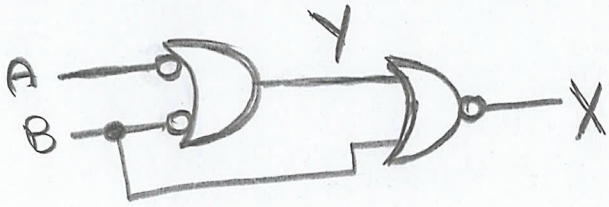


NOR

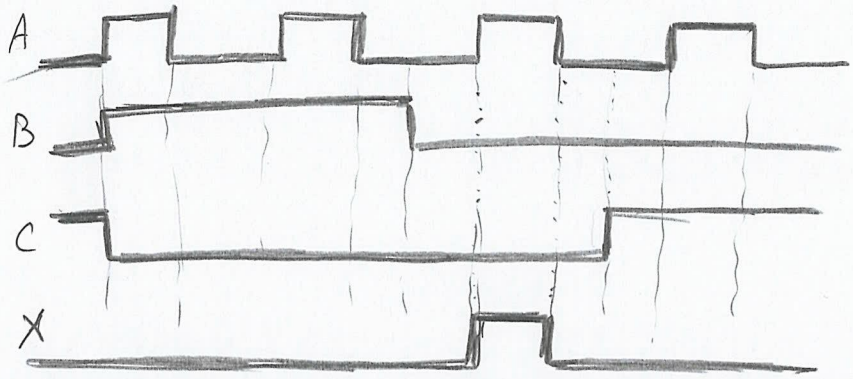


28





31



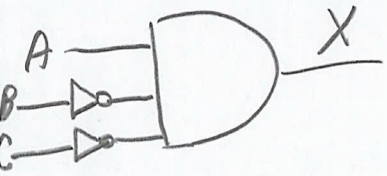
⇒

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	X

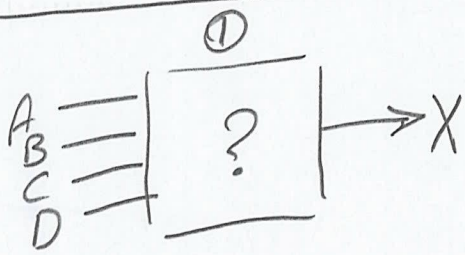
⇓

AB \ C	0	1
00		
01		X
11		X
10	1	

⇐ $X = A\bar{B}\bar{C}$ ⇐



56



⇒

2

A	B	C	D	X
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

<3 ↑

↓ >12

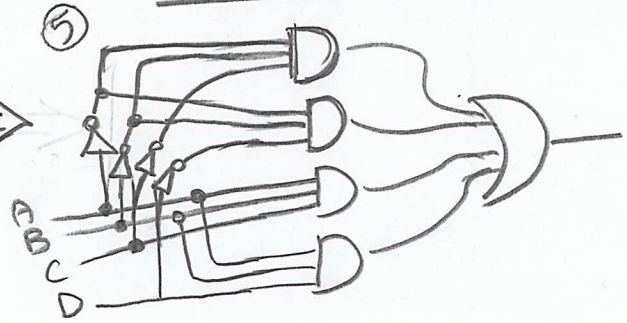
⇐

3

AB \ CD	00	01	11	10
00	1	1		1
01				
11		1	1	1
10				

⇓

4

$X = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{D} + ABC + ABD$ ⇒

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